Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 5 — Solutions

October 9, 2026

Exercise 1. *Let A and B be sets. For each of the following cases, determine whether the given object defines a relation from A to B. Which of the following statements are correct?*

- ✓ *Let $A = \{1, 2, 3\}$, $B = \{a, b\}$, and $R = \{(1, a), (2, b), (3, a)\}$. Then R is a relation from A to B.*
- ◯ *Let $A = \{1, 2, 3\}$, $B = \{a, b\}$, and $R = \{(1, a), (2, b), (4, a)\}$. Then R is a relation from A to B.*
- ✓ *Let $A = \mathbb{Z}$, $B = \mathbb{Z}$, and $R = \{(x, y) \in \mathbb{Z}^2 : y = 0\}$. Then R is a relation from A to B.*
- ◯ *Let $A = \mathbb{R}$, $B = \mathbb{R}$, and $R = \{x \in \mathbb{R} : x^2 < 1\}$. Then R is a relation from A to B.*

Recall that a relation from A to B is any subset $R \subseteq A \times B$, i.e. a set of ordered pairs (a, b) with $a \in A$ and $b \in B$.

- **True.** Every pair of R has its first component in A and its second in B, so $R \subseteq A \times B$.
- **False.** $(4, a) \in R$ but $4 \notin A$, so $(4, a) \notin A \times B$ and $R \not\subseteq A \times B$.
- **True.** Every element of R is of the form $(x, 0)$ with $x \in \mathbb{Z}$ and $0 \in \mathbb{Z}$, so $R \subseteq \mathbb{Z} \times \mathbb{Z}$: R *is* a relation.
- **False.** R is a set of real numbers, not of ordered pairs, so $R \not\subseteq \mathbb{R} \times \mathbb{R}$. (By contrast, $\{(x, y) \in \mathbb{R}^2 : x^2 < 1\}$ would be a relation.)

Therefore, the only correct statement is the first one.

Exercise 2. *For each relation below, determine which of the following properties it satisfies: reflexive, symmetric, anti-symmetric, transitive.*

(a) *On $\mathbb{Z}$, $xRy \iff x - y$ is even.*
Reflexive, symmetric and transitive: $x - x = 0$ is even; if $x - y$ is even, so is $y - x = -(x - y)$; if $x - y$ and $y - z$ are even, so is $x - z = (x - y) + (y - z)$.
Not anti-symmetric: $0R2$ and $2R0$ but $0 \neq 2$.

(b) *On $\mathbb{Z}$, $xRy \iff x < y$.*
Anti-symmetric (vacuously, $x < y$ and $y < x$ never hold simultaneously) and transitive ($x < y < z \Rightarrow x < z$).
Not reflexive: $0 \not< 0$. Not symmetric: $0 < 1$ but $1 \not< 0$.

(c) *On $\mathbb{Z}$, $xRy \iff x \leq y$.*
Reflexive ($x \leq x$), anti-symmetric ($x \leq y \wedge y \leq x \Rightarrow x = y$) and transitive ($x \leq y \leq z \Rightarrow x \leq z$).
Not symmetric: $0 \leq 1$ but $1 \not\leq 0$.

(d) *On $\mathbb{R}$, $xRy \iff |x - y| \leq 1$.*
Reflexive ($|x - x| = 0 \leq 1$) and symmetric ($|x - y| = |y - x|$).
Not anti-symmetric: $0R1$ and $1R0$ but $0 \neq 1$. Not transitive: $0R1$ and $1R2$ but $|0 - 2| = 2 > 1$.

(e) *On $\mathbb{Z}$, $xRy \iff xy \geq 0$.*
Reflexive ($x^2 \geq 0$) and symmetric ($xy = yx$).
Not anti-symmetric: $1R2$ and $2R1$ but $1 \neq 2$. Not transitive: $1R0$ and $0R(-1)$ but $1 \cdot (-1) = -1 < 0$.

(f) *On* $\mathbb{N}$, $xRy \iff x \mid y$.
Reflexive ($x = 1 \cdot x$), transitive (if $y = kx$ and $z = \ell y$, then $z = (k\ell)x$) and anti-symmetric: if $y = kx$ and $x = \ell y$, then $x = k\ell x$, so either $x = 0$ (and then $y = 0 = x$) or $k\ell = 1$, which forces $k = \ell = 1$ in $\mathbb{N}$, hence $x = y$.
Not symmetric: $1 \mid 2$ but $2 \nmid 1$.

Exercise 3. *Let* $A = \{1, 2, 3, 4\}$ *and* $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (3, 4)\}$.

1. *Which of the equivalence-relation properties fail for* R*?*
2. *We want to add the minimum possible number* n *of ordered pairs to* R *so that it becomes an equivalence relation. Determine* n *and pair(s) that should be added.*

1. **Reflexivity fails:** $(4, 4) \notin R$.
Symmetry fails: $(3, 4) \in R$ but $(4, 3) \notin R$.
Transitivity holds
2. $n = 2$: we add $(4, 4)$ and $(4, 3)$.
Both are necessary: $(4, 4)$ is required by reflexivity and $(4, 3)$ by symmetry, so $n \geq 2$.
They are sufficient.

Exercise 4. *A binary relation* R *on a set* A *is said to be quasi-reflexive if every element that is part of the relation is related to itself:* $\forall a \in A$ *and* $\forall b \in A$*, we have* $(a, b) \in R \to [(a, a) \in R \wedge (b, b) \in R)]$*. Let us assume that we have a non-empty binary relation* R *on a set* A *that is both symmetric and anti-symmetric. Then* R *is:*

◯ *Always reflexive and always quasi-reflexive*

◯ *Always reflexive but not always quasi-reflexive*

✓ *Not always reflexive but always quasi-reflexive*

◯ *Not always reflexive and not always quasi-reflexive*

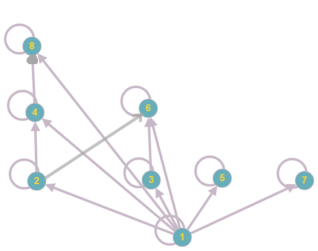
R is both symmetric and anti-symmetric.
Therefore elements of R must follow the two constraints:

- $(a, b) \in R \to (b, a) \in R$
- $[(a, b) \in R \wedge (b, a) \in R] \to a = b$

This implies that R contains only elements that are related to themselves, indeed if $(a, b) \in R$, then $(b, a) \in R$ because R is symmetric and this implies that $a = b$ because R is anti-symmetric.
To sum up elements of R follow the following property: $(a, b) \in R \to a = b$.
We can conclude that R is quasi-reflexive because $(a, b) \in R \to a = b \to [(a, a) \in R \wedge (b, b) \in R)]$.
If we take $A = \{0, 1\}$ and $R = \{(0, 0)\}$, R is symmetric, anti-symmetric, quasi-reflexive but not reflexive.
Therefore, the correct answer is: *"Not always reflexive but always quasi-reflexive"*.

Exercise 5. *Draw the Hasse diagram for divisibility on the set:*

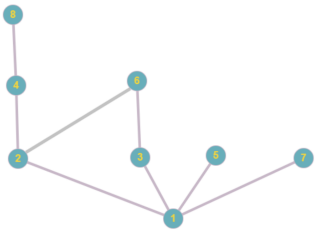
1. *{1, 2, 3, 4, 5, 6, 7, 8}*



(a)



(b)



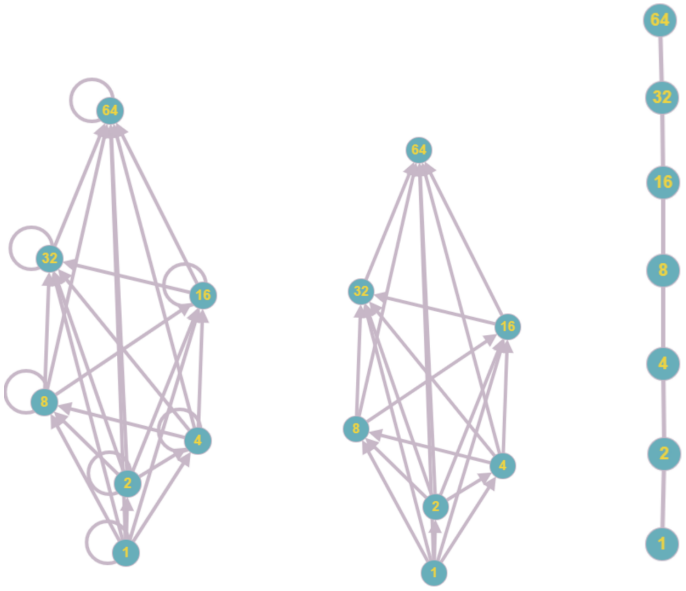
(c)

2. $\{1, 2, 3, 5, 7, 11, 13\}$



(a) (b) (c)

3. $\{1, 2, 4, 8, 16, 32, 64\}$



(a) (b) (c)

Exercise 6. *Suppose that* $(S, \preceq_1)$ *and* $(T, \preceq_2)$ *are posets. Show that* $(S \times T, \preceq)$ *is a poset where* $(s, t) \preceq (u, v)$ *if and only if* $s \preceq_1 u$ *and* $t \preceq_2 v$.

PROOF $(S \times T, \preceq)$ is a poset if and only if the relation $R = \{((s, t), (u, v))|(s, t) \preceq (u, v)\}$ is a partial ordering.

$(S, \preceq_1)$ and $(T, \preceq_2)$ are posets, thus the relations $R_1 = \{(s, u)|s \preceq_1 u\}$ and $R_2 = \{(t, v)|t \preceq_2 v\}$ are both reflexive, antisymmetric and transitive.

Reflexive
Let $(s, t) \in S \times T$, where $s \in S$ and $t \in T$.
Since $R_1 = \{(s, u)|s \preceq_1 u\}$ and $R_2 = \{(t, v)|t \preceq_1 v\}$ are both reflexive:

$$s \preceq_1 s$$
$$t \preceq_2 t$$

$(s, t) \preceq (u, v)$ if and only if $s \preceq_1 u$ and $t \preceq_2 v$

$$(s, t) \preceq (s, t)$$

which implies $((s, t), (s, t)) \in R$ and thus R is reflexive.

Antisymmetric
Let $((s, t), (u, v)) \in R$ and $((u, v), (s, t)) \in R$

$$(s,t) \preceq (u,v)$$
$$(u,v) \preceq (s,t)$$

$(s,t) \preceq (u,v)$ if and only if $s \preceq_1 u$ and $t \preceq_2 v$

$$s \preceq_1 u$$
$$u \preceq_1 s$$
$$t \preceq_2 v$$
$$v \preceq_2 t$$

Since $R_1 = \{(s,u) | s \preceq_1 u\}$ and $R_2 = \{(t,v) | t \preceq_2 v\}$ are both antisymmetric:

$$s = u,\ t = v$$

which implies:

$$(s,t) = (u,v)$$

Thus R is antisymmetric.

Transitive
Let $((s,t),(u,v)) \in R$ and $((u,v),(w,x)) \in R$

$$(s,t) \preceq (u,v)$$
$$(u,v) \preceq (w,x)$$

$(s,t) \preceq (u,v)$ if and only if $s \preceq_1 u$ and $t \preceq_2 v$

$$s \preceq_1 u$$
$$u \preceq_1 w$$
$$t \preceq_2 v$$
$$v \preceq_2 x$$

Since $R_1 = \{(s,u) | s \preceq_1 u\}$ and $R_2 = \{(t,v) | t \preceq_2 v\}$ are both transitive:

$$s \preceq_1 w$$
$$t \preceq_2 x$$

which implies:

$$(s,t) \preceq (w,x)$$

Thus R is transitive.
Conclusion: R is reflexive, antisymmetric and transitive. Then R is a partial ordering and $(S \times T, R)$ is a poset.

Exercise 7. *Find the lexicographic ordering of these n-tuples:*

1. $(1,1,2) \prec (1,2,1)$
2. $(0,1,2,3) \prec (0,1,3,2)$
3. $(0,1,1,1,0) \prec (1,0,1,0,1)$

Exercise 8. *For each of these sequences find a recurrence relation satisfied by this sequence. (The answers are not unique because there are infinitely many different recurrence relations satisfied by any sequence.)*

1. Sequence: $a_n = 3$
   $a_0 = 3$ and $a_n = a_{n-1}$
2. Sequence: $a_n = 2n$
   $a_0 = 0$ and $a_n = a_{n-1} + 2$

3. Sequence: $a_n = 2n + 3$
   $a_0 = 3$ and $a_n = a_{n-1} + 2$

4. Sequence: $a_n = 5^n$
   $a_0 = 1$ and $a_n = 5a_{n-1}$

5. Sequence: $a_n = n^2$
   $a_0 = 0$ and $a_n = a_{n-1} + 2(n-1) + 1$

6. Sequence: $a_n = n^2 + n$
   $a_0 = 0$ and $a_n = a_{n-1} + 2(n-1) + 2$

7. Sequence: $a_n = n + (-1)^n$
   $a_0 = 1$ and $a_n = a_{n-1} + 1 + 2(-1)^n$

8. Sequence: $a_n = n!$
   $a_0 = 1$ and $a_n = na_{n-1}$

Exercise 9. *Find the solution to each of these recurrence relations and initial conditions.*

1. Recurrence relations and initial conditions: $a_n = 3a_{n-1}$ and $a_0 = 2$
   $a_n = 3 \cdot a_{n-1} = 3 \cdot 3 \cdot a_{n-2} = \underbrace{3 \cdot 3 \cdot ... \cdot 3}_{3^n} \cdot \underbrace{a_0}_{2} = 3^n \cdot 2$

2. Recurrence relations and initial conditions: $a_n = a_{n-1} + 2$ and $a_0 = 3$
   $a_n = 2 + a_{n-1} = 2 + 2 + a_{n-2} = \underbrace{2 + 2... + 2}_{2n} + \underbrace{a_0}_{3} = 2n + 3$

3. Recurrence relations and initial conditions: $a_n = a_{n-1} + n$ and $a_0 = 1$
   $a_n = n + a_{n-1} = n + (n-1) + a_{n-2} = \underbrace{n + (n-1) + ... + 1}_{\sum_{i=1}^{n} i = (n^2+n)/2} + \underbrace{a_0}_{1} = (n^2 + n)/2 + 1$

4. Recurrence relations and initial conditions: $a_n = a_{n-1} + 2n + 3$ and $a_0 = 4$
   $a_n = 2n + 3 + a_{n-1} = 2n + 3 + 2(n-1) + 3 + ... + 2 + 3 + \underbrace{a_0}_{4} = 2(n^2 + n)/2 + 3n + 4 = n^2 + 4n + 4$

5. Recurrence relations and initial conditions: $a_n = 2a_{n-1} - 1$ and $a_0 = 1$
   $a_n = 2a_{n-1} - 1 = 2(2a_{n-2} - 1) - 1 = 2(2(2... \underbrace{2(\underbrace{a_0}_{1} - 1)}_{1} - 1) - ... - 1) - 1) - 1 = 1$

6. Recurrence relations and initial conditions: $a_n = 3a_{n-1} + 1$ and $a_0 = 1$
   $a_n = 3a_{n-1} + 1 = 3^2 a_{n-2} + 3 + 1 = 3^3 a_{n-3} + 3^2 + 3 + 1 = 3^n \underbrace{a_0}_{1} + ... + 3 + 1 = \sum_{k=0}^{n} 3^k = \frac{3^{n+1}-1}{2}$

7. Recurrence relations and initial conditions: $a_n = na_{n-1}$ and $a_0 = 5$
   $a_n = na_{n-1} = n(n-1)a_{n-2} = n(n-1)...(1) \underbrace{a_0}_{5} = 5n!$

8. Recurrence relations and initial conditions: $a_n = 2na_{n-1}$ and $a_0 = 1$
   $a_n = 2na_{n-1} = 2n(2(n-1)a_{n-2}) = 2n2(n-1)...2(1) \underbrace{a_0}_{1} = 2^n n!$

Exercise 10. *Suppose that the number of bacteria in a colony triples every hour.*

1. *Set up a recurrence relation for the number of bacteria after n hours have elapsed*
   Let a_n represents the number of bacteria after n hours have elapsed. Every hour, the number of bacteria triples. Thus the number of bacteria is the number of bacteria at an hour ago multiplied by 3.

$$a_n = 3a_{n-1}$$

2. *If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 hours?*
   Given:

$$\begin{gathered} a_n = 3a_{n-1} \\ a_0 = 100 \end{gathered}$$

   We successively apply the recurrence relation:

$$\begin{gathered} a_n = 3a_{n-1} = 3^1 a_{n-1} \\ = 3(3a_{n-2}) = 3^2 a_{n-2} \\ = 3^2(3a_{n-3}) = 3^3 a_{n-3} \\ = 3^3(3a_{n-4}) = 3^4 a_{n-4} \\ \cdots \\ = 3^n a_{n-n} \\ = 3^n a_0 \\ = 100 \cdot 3^n \end{gathered}$$

   Evaluate the found expression at $n = 10$:

$$a_{10} = 100 \cdot 3^{10} \approx 5,904,900$$

   Thus there are 5,904,900 bacteria after 10 hours.

Exercise 11. *Solve the following recurrence relation using generating functions:* $a_n = 4a_{n-1} - 4a_{n-2}$, $n \geq 2$, *with* $a_0 = 1$ *and* $a_1 = 4$.

Multiplying by x^n and summing over all $n \geq 2$ we get

$$\sum_{n\geq 2} a_n x^n = 4\sum_{n\geq 2} a_{n-1}x^n - 4\sum_{n\geq 2} a_{n-2}x^n.$$

Rearranging, reindexing, and letting $A(x) = \sum_{n\geq 0} a_n x^n$ we get

$$A(x) - a_0 - a_1 x = 4x(A(x) - a_0) - 4x^2 A(x).$$

This has the solution

$$A(x) = \frac{a_0 + a_1 x - 4xa_0}{1 - 4x + 4x^2} = \frac{1}{(1-2x)^2}.$$

This already has the form of the partial fraction and note that we have a double root. From the lecture we know that this has the expansion

$$A(x) = \frac{1}{(1-2x)^2} = \sum_{n\geq 0}(n+1)2^n x^n.$$

It is always a good idea to make a sanity check and to plug this back into the recursion. We get

$$(n+1)2^n = 4n2^{n-1} - 4(n-1)2^{n-2}.$$

Dividing by 2^n we get

$$(n+1) = 2n - (n-1) = (n+1)$$

Exercise 12. *We have seen in the course that we can add, multiply, and, if the divisor has a non-zero constant term, divide formal power sums (FPS). In addition we have seen that we can take the formal derivative of a FPS. The key for all those operations is that we can compute the n-th coefficient of the result using only a finite number of basic operations. There are many other operations one can define.*

Let $A(x) = \sum_{n\geq 0} a_n x^n$ and $B(x) = \sum_{n\geq 0} b_n x^n$ be two FPS. Under what condition is the composition $A(B(x))$ well defined? Compute the first 3 coefficients.

We want

$$A(B(x)) = a_0 + a_1 B(x) + a_2 B(x)^2 + a_3 B(x)^3 + \cdots$$

To find, say, the constant term of this expression, we need to add up the constant terms of a_0, $a_1 B(x)$, $a_2 B(x)^2$, $a_3 B(x)^3$, and so on – for *every* n, not just the first few. Let's see what that gives us.

Write $B(x) = b_0 + b_1 x + b_2 x^2 + \cdots$. If we multiply $B(x)$ by itself a few times, we get

$$\begin{aligned} B(x)^2 &= (b_0 + b_1 x + b_2 x^2 + \cdots)(b_0 + b_1 x + b_2 x^2 + \cdots) \\ &= b_0^2 + 2b_0 b_1\, x + (2b_0 b_2 + b_1^2)\, x^2 + \cdots \\ B(x)^3 &= B(x)^2 \cdot B(x) = b_0^3 + 3b_0^2 b_1\, x + \cdots \end{aligned}$$

and in general, whenever $b_0 \neq 0$, the constant term of $B(x)^n$ is b_0^n, which is never 0. So every single power $B(x)^n$, for every $n = 0, 1, 2, 3, \ldots$, contributes something to the constant term of $A(B(x))$.

That means the constant term of $A(B(x))$ would have to be

$$a_0 + a_1 b_0 + a_2 b_0^2 + a_3 b_0^3 + \cdots,$$

a sum with infinitely many terms. This is different from what happens with addition, multiplication, division or differentiation, where each coefficient of the answer only ever involves a handful of the a_i's and b_j's. Here we would need to add up infinitely many numbers just to get one single coefficient, and there is no reason this should make sense: we are not allowed to take limits or talk about convergence for formal power sums, we can only add finitely many numbers at a time. So if $b_0 \neq 0$, the coefficients of $A(B(x))$ simply cannot be computed, and the same problem shows up for the coefficient of x^1, x^2, x^3, etc.

So the composition $A(B(x))$ does not make sense in general when $b_0 \neq 0$.

The whole issue came from one thing: $B(x)^n$ has a nonzero constant term for every n. This will stop happening if we require $b_0 = 0$, i.e.

$$B(x) = b_1 x + b_2 x^2 + b_3 x^3 + \cdots.$$

Let's check why this helps. If $b_0 = 0$, every term of $B(x)$ contains at least one factor of x. So when we multiply $B(x)$ by itself n times to get $B(x)^n$, every single term in the result is a product of n things that each have at least one factor of x – so every term of $B(x)^n$ has at least x^n in it. In other words,

$$B(x)^n = (\text{something}) \cdot x^n,$$

so $B(x)^n$ cannot contribute anything to the coefficients of $1, x, x^2, \ldots, x^{n-1}$: those coefficients are all 0.

This is exactly what we need: to compute the coefficient of x^k in $A(B(x))$, we only need to look at the terms $a_0, a_1 B(x), \ldots, a_k B(x)^k$, because every later term $a_n B(x)^n$ with $n > k$ does not contribute to x^k at all (its lowest power of x is x^n, which is higher than x^k). So now we are back to a finite computation.

Conclusion: $A(B(x))$ is well defined exactly when $b_0 = 0$.

Now assume $b_0 = 0$, so $B(x) = b_1 x + b_2 x^2 + b_3 x^3 + \cdots$. Let's compute $B(x)^2$ and $B(x)^3$ again, but now keeping only the terms up to x^3 (we don't need anything beyond that):

$$\begin{aligned} B(x)^2 &= (b_1 x + b_2 x^2 + b_3 x^3 + \cdots)^2 = b_1^2 x^2 + 2b_1 b_2\, x^3 + \cdots \\ B(x)^3 &= (b_1 x + b_2 x^2 + \cdots)^3 = b_1^3 x^3 + \cdots \end{aligned}$$

(For $B(x)^2$, to get an x^2 term we must pick x from both factors, giving $b_1 \cdot b_1$; to get an x^3 term we pick x from one factor and x^2 from the other, in 2 ways, giving $2b_1b_2$. For $B(x)^3$, the lowest power we can possibly get is x^3, by picking x from each of the three factors, giving b_1^3.)

Now we just add up, for each power of x, the contributions coming from $a_0, a_1B(x), a_2B(x)^2, a_3B(x)^3$:

- constant term: only a_0 contributes: a_0
- coefficient of x: only $a_1B(x)$ contributes: a_1b_1
- coefficient of x^2: $a_1B(x)$ and $a_2B(x)^2$ contribute: $a_1b_2 + a_2b_1^2$
- coefficient of x^3: $a_1B(x)$, $a_2B(x)^2$ and $a_3B(x)^3$ contribute: $a_1b_3 + 2a_2b_1b_2 + a_3b_1^3$

So altogether:

$$A(B(x)) = a_0 + a_1b_1\,x + (a_1b_2 + a_2b_1^2)\,x^2 + (a_1b_3 + 2a_2b_1b_2 + a_3b_1^3)\,x^3 + \cdots$$